

Problem 1: The Last Astronaut

Pseudocode

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INPUT N, K
CREATE array eliminated[N], isi semua false
pos = 0
remaining = N

WHILE remaining > 1
    count = 0
    idx = pos
    WHILE count < K
        IF eliminated[idx] == false
            count = count + 1
        IF count == K
            BREAK
        idx = (idx + 1) MOD N

    PRINT "Astronot " + (idx + 1) + " dieliminasi"
    eliminated[idx] = true

    IF (idx + 1) MOD 2 == 0
        K = K + 2
    ELSE
        K = K - 1
    IF K < 2
        K = 2

    next = (idx + 1) MOD N
    WHILE eliminated[next] == true
        next = (next + 1) MOD N
    pos = next

    remaining = remaining - 1

FOR i = 0 TO N - 1
    IF eliminated[i] == false
        PRINT "Astronot terakhir yang bertahan: " + (i + 1)
```

Problem 2: Alien In The Middle

Flowchart

